

Smart Waiter Agent Workflows

This document outlines the core workflows of the Smart Waiter AI Agent. The agent is built using **LangGraph** and operates as a state machine that routes user inputs to specific sub-processes based on intent.

High-Level Architecture

The **SmartWaiterAgent** (`agents/smart_waiter_agent.py`) acts as the central brain.

- 1. **Input**: User sends a message (Text or Voice).
- 2. **Table**

1. Onboarding Workflow

File: `workflows/onboarding_flow.py`

Ensures essential session data is collected before service begins.

```
graph TD
    Start([User Starts]) --> Decider{Missing Info?}
    Decider -->|Language?| AskLang[Ask Language]
    Decider -->|Mode?| AskMode[Ask Voice/Text]
    Decider -->|Table?| AskTable[Ask Table #]

    AskLang -->|User Replies| Decider
    AskMode -->|User Replies| Decider
    AskTable -->|User Replies| Decider

    Decider -->|All Set| End([Onboarding Complete])
```

- **Languages Supported:** English, Hausa, Yoruba, Igbo, French.
- **P**

Smart Waiter System Workflows

2. Ordering Workflow

File: `workflows/ordering\_flow.py`

Handles menu inquiries and adding items to the cart.

```
graph LR
    Input([User Request]) --> Interpret[LLM Interpretation]
    Interpret -->|JSON Action| Execute[Execute Tool]

    Execute -->|Add Item| AddOps[order_ops.add_item]
    Execute -->|Remove Item| RemOps[order_ops.remove_item]
    Execute -->|None/Invalid| NoOp

    AddOps --> Upsell{Category Check}
    Upsell -->|Main Dish?| OfferProtein[Upsell Meat/Fish]
    Upsell -->|Other| Recommend[Recommend Drink/Side]

    OfferProtein --> Summary[Generate Order Summary]
    Recommend --> Summary
    RemOps --> Summary

    Summary --> Respond([Final Response])
```

- **Safety:** strictly validates items against `menu\_ops`.

- **Lo**

3. Payment Workflow

File: `workflows/payment\_flow.py`

Manages checkout, method selection, verification, and closing the table.

```
graph TD
    Start([Payment Intent]) --> Analyze[Current Status]

    Analyze -->|Init| ShowBill[Itemise Bill & Ask Method]
    ShowBill --> WaitUser

    Analyze -->|Method Selected| Process[Process Method]

    Process -->|Transfer| ShowBank[Show Bank Details]
    Process -->|Card/Cash| AlertAdmin[Alert Admin & Wait]

    ShowBank -->|User confirms| AskDetails[Ask Name/Amount]
    AskDetails -->|Details Given| AlertAdmin
```

Smart Waiter System Workflows

```
AlertAdmin -->|Admin Click Verify| Paid[Paid Success]
AlertAdmin -->|Admin Click Deny| Fail[Payment Failed]

Paid --> AskDisp[Ask: Eat In or Takeaway?]
AskDisp -->|User Replies| Close[Send Ticket to Kitchen]
```

- **Integration:** Sends real-time alerts to the **Admin Telegram Channel** with "Approve/Deny" buttons.

4. Fulfillment & Feedback Workflow

File: `workflows/fulfillment\_flow.py`

Handles post-payment interactions.

1. **Status Check:** User asks "Where is my food?". Agent checks `kitchen\_ops`.

Admin Interaction

File: `telegram\_bot.py` (Callback Handlers)

- **Approvals:** Admin receives a message with `Approve` / `Deny` buttons.